

Services as Activities: Towards a Unified Definition for (Public) Services

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Abstract— I present a novel definition of services (in particular, *public services*) which sees each of them as the sum of all *activities* that realize a public authority's *commitment* to make *available* to individuals, businesses, or other public authorities some *capabilities* intended to answer their *needs*, giving them some possibilities to *control* how and when such capabilities should be manifested.

Keywords—*services definition*

I. INTRODUCTION

In a recent paper [1], Nardi and colleagues considered different ways of characterizing services, and proposed an ontology that explicates and harmonizes the different perspectives. They analyzed the basic phases of the service lifecycle, namely *offering*, *negotiation*, and *delivery*, emphasizing the different role of *relationships* between service providers and customers in each of such phases, and showing how the notions belonging to the various service perspectives can be mapped to such ontology. They refrained however from proposing a specific definition for the term 'service', considering it as a case of *systematic polysemy*, in which the same name is used to denote different –although related– notions.

This is certainly true, but still there is the urgent practical need of reducing this polysemy as much as possible, trying to capture the core notions that underlie the way we talk of services in our everyday language. In Europe, this need emerges in particular for *public* services, which in principle should be accessible to all European citizens in a homogeneous way, but in practice reflect the idiosyncrasies of the various public administrations. For instance, according to the 2017 ISA European workprogramme for the semantic interoperability of public administrations, there is still no standard way of describing and documenting European public services:

“Different types of approaches are being adopted towards the building of public services and their catalogues. There is usually no standard way of describing and documenting these services. The understanding of services and service implementations are different and *even the basic definition of what constitutes a public service differs*. Furthermore, there is a lack of an overview of what types of services already exist, often resulting in redundant work and inefficiencies”. ([7], p. 26, our emphasis)

On the other hand, an important strategic document such as the European Services Directive adopts a relatively simple definition of (economic) services, namely:

" 'service' means any self-employed economic *activity*, normally provided for remuneration" ([8], Article 4, our emphasis)

Another definition, oriented towards public services, is reported in the recent “Study of the new generation of eGovernment Services”¹:

"Public services are *activities* that are publicly funded and arise from public policy and that are for the collective benefit of the public, accountable to and governed by a political process. This includes both administrative and human services." (our emphasis)

Both these definitions suggest that services are *activities* of a certain kind, although the exact nature of such activities is not clarified. In this paper we go back to the original challenge – coming up with a *unifying* service definition– by presenting an ontological account of *services as activities*. Such proposal leverages on the early foundational work by Ferrario and Guarino [2], which sees a service as a complex temporal entity consisting of a *service commitment* and the corresponding process. While in that work (and in the follow-up work with Nardi and colleagues [1]) we stressed the role of service commitment, considering its mere existence as sufficient for the existence of a service, here we focus our attention on other aspects of services, such as *capability*, *availability*, and *control*. As we shall see, *all* these will become essential ingredients of our service definition.

¹ https://joinup.ec.europa.eu/community/opengov/og_page/ogs-study

² <https://joinup.ec.europa.eu/asset/cpsv-ap/description>

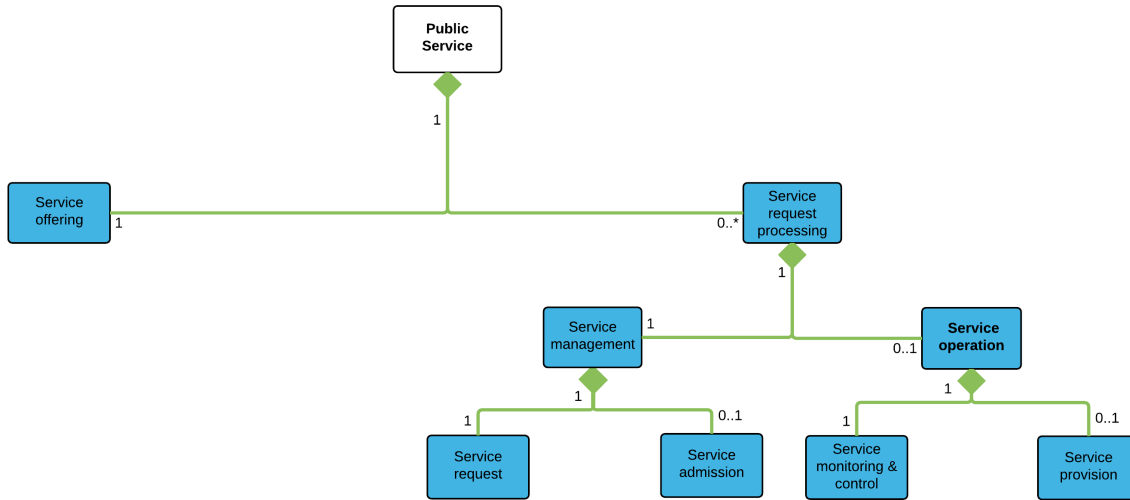


Figure 1. Public service as a sum of activities

II. SERVICES AS ACTIVITIES: CLARIFYING THE AMBIGUITIES

According to Ferrario and Guarino, behind every service there is a commitment to perform *actions* of a certain kind. For instance, behind a mobile phone service there is a commitment to perform phone connection actions. Of course, at a *microscopic* level of analysis, each of such actions *may* be considered as a service, but this is not the notion of service people use in their everyday speaking: people subscribe to *one* telephone service, which offers them the possibility to activate multiple phone connections. Each of these connections is part – so to speak – of *the same* service. This means that, while analyzing services, we should first of all distinguish *core service actions* from *activities* involving such actions. We shall say that a *service provision* is an activity resulting from the aggregation of multiple service actions for the benefit of a single customer. In turn, a *service* is an aggregation of service provisions, possibly involving multiple customers. In this way, we can have a single entry in a service catalog that refers to a unique service, independently of the number of actual service provisions and the number of core service actions involved in each of such provisions. In other words, to capture the everyday notion of service we need to move from the microscopic to the *mesoscopic level of analysis* [3], at which multiple service actions of the same kind are aggregated in a coherent whole. It is such coherent whole that, in our opinion, deserves to be called a service.

To better understand the issue, consider the definition of public service proposed by the latest version of the Core Public Service Vocabulary Application Profile (CPSV-AP)²:

"A Public Service is a *mandatory or discretionary set of acts* performed, or able to be performed, by or on behalf of a public organization. Services may be for the benefit of an individual, a business, or other public authority, or groups of any of these. The capacity to act exists whether it is used or not, and the term 'benefit' may apply in the sense of enabling the fulfillment of an obligation." (our emphasis)

This definition is still in agreement with the previous ones (since an activity can be understood as a set of acts), with the

difference that here the nature of such acts is clarified a bit more. Unfortunately, however, it leaves open whether the single acts belonging to the set that constitutes a public service concern a single *service provision*, for the benefit of a single customer, or rather a single *service offering*, addressed to multiple customers. Consider for instance a public nursery service: each service provision addressed to a parent or a family consists of a sequence of acts performed by a public organization for the benefit of an individual, or a group of individuals (the family). So, a single service provision would count as a public service according to this definition. In this case, a public service identifier may be the registration number formally issued once the nursery request is processed.

On the other hand, a single service provision is probably not the kind of 'Public service' we have in mind: indeed, especially from the point of view of a service catalog, what we need is being able to list public services described by a single *service offering*, addressed to multiple customers. Note that a service understood in this way would still satisfy the definition above, since it may be seen as a (large) set of acts performed by a public organization for the benefit of an individual. However, in this case the service identifier would be rather different, since it should be somehow attached to the service offering.

We can conclude that the CPSV-AP definition, as well as the previous definitions of services as activities, fails in offering a way to identify and count services, because of the ambiguity between *service* and *service provision*. A way to solve this ambiguity, accounting for the different nature of these two notions and their mutual relationships is described in Fig. 1. A public service, intended as a sum of acts (and therefore as an activity), is composed in turn by two main kinds of activities: a *service offering* activity and a number of *service request processing* activities, one for each service request. In turn, service request processing includes *service management* activities, such as those needed to handle *service requests* and decide about *service admission*, and *service operation* activities, such as the actual *service provision* and the

² <https://joinup.ec.europa.eu/asset/cpsv-ap/description>

capabilities are manifested.

Note that this definition marks a radical difference with respect to the previous definition proposed by Ferrario and Guarino, since in this case the commitment is not *part* of the service, although the service still depends on it. Moreover, now the commitment does not concern directly the execution of core service actions, but rather the availability of certain capabilities. In this way we can see the actions done to make the service available to customers, including those concerning the offering phase, as proper parts of the service. In this way, all the various aspects of services described separately by Nardi and colleagues [1], namely *service offering*, *service negotiation*, and *service delivery* can be seen as *parts* of what a service is.

Before showing briefly how this definition can be applied in practice, let me make a couple of comments concerning the ontological choices behind the analysis presented here. A first comment is about the ontological nature of activities. Of course, an option is to consider them as temporal occurrences (a.k.a. *perdurants* in the formal ontology literature). In this case, they would correspond to what in DOLCE [9] are called *processes*. This choice is however not particularly amenable for our purposes, since temporal occurrences in DOLCE are entities frozen in time, which cannot change. This would force us to model services only as historical entities, after they have ceased to be present. On the contrary, the idea of services as activities relies on the intuition that such activities are typically *ongoing*, and can genuinely change in time by acquiring new temporal parts. This requires a non-standard view of temporal occurrences, defended in a recent paper [5].

A second comment concerns the connection between this approach and the one presented by Nardi and colleagues, which is mainly focused on *service relationships*. Why don't relationships occur in our new model? The answer is that they are implicitly there, as the *focus* of the various activities shown in the model (see [6] for a recent discussion on the interplay of events and relationships).

IV. SPECIALIZING THE CPSV-AP

Once the core pattern of Fig. 1 is adopted, together with the definition above, we can describe different kinds of public services in an accurate way, clarifying the agents involved in a certain activity kind, as well as the service channels and the descriptions attached to the various activities. Fig. 2 shows a possible extension of the CPSV-AP based on the view discussed here. The original CPSV-AP classes are in white, the various service activities are in blue, service roles (such as *service authority*, *provider*, *customer*, and *consumer*) are in red, service descriptions and rules are in yellow, while the various service communication channels are in brown. The graph shows how specific aspects of a service concern specific parts of it, in the sense discussed here. So, for instance, a complex service offering description is attached to the service offering activity. Such description incorporates a service provision description as a part. Note that no service negotiation activity appears in the graph, since a negotiation phase is typically present in business services but not in public services. Rather, the service management activity includes the service admission activity, which typically has to comply with specific rules. The graph shows also how the various service activities include specific activities devoted to the communication with the customers, called *channels*. Different channels are foreseen for different service activities.

This graph is only indicative of the way the CPSV-AP might be extended at a rather fine-grained level but still in a conservative way, that is, keeping the current definitions for all classes except the one Public Service, which is revisited according to the proposal we have discussed here.

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